Computational Marine Ecological Modelling (25314)

NPZDO model

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List of Symbols

Symbol	Unit	Description
C	varies	Generic tracer concentration
D	$\mathrm{mmol\,N\,m^{-3}}$	Detrital organic nitrogen concentration in the water column
D_{n}	$\mathrm{mmol\,N\,m^{-3}}$	Detritus concentration in the bottom water cell
D_{sed}	$\mathrm{mmol\,N\,m^{-2}}$	Sediment detritus inventory
$F_{\mathrm{air-sea}}$	$\mathrm{mmol\,O_2\,m^{-2}\,d^{-1}}$	Air–sea oxygen exchange flux
F_{ben}	$\mathrm{mmol\,N\,m^{-2}\,d^{-1}}$	Benthic remineralization flux
F_{dep}	$\mathrm{mmol\,N\,m^{-2}\,d^{-1}}$	Deposition flux of detritus from the water column to the sediment
F_{graz}	$\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$	Grazing flux
F_{mortP}	$\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$	Phytoplankton mortality flux
F_{mortZ}	$\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$	Zooplankton mortality flux
F_{prod}	$\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$	Primary production flux
F_{relin}	$\mathrm{mmol\,N\,m^{-2}\,d^{-1}}$	Sediment nutrient release flux
F_{remin}	$\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$	Water-column remineralization flux
F_{resp}	$\mathrm{mmol\,N\,m^{-3}\,d^{-1}}$	Respiration and excretion flux returning nitrogen to dissolved nutrients
G_{lim}	–	Grazing limitation factor
g_{max}	$\mathrm{d^{-1}}$	Maximum zooplankton grazing rate
J	varies	Vertical tracer flux
k_L	$\mathrm{\mu mol\,m^{-2}\,s^{-1}}$	Light half-saturation constant
k_N	$\mathrm{mmol\,N\,m^{-3}}$	Nutrient half-saturation constant
$k_{N,\mathrm{n}}$	$\mathrm{m\,d^{-1}}$	Sediment–water nitrogen exchange velocity
k_Z	$\mathrm{mmol\,N\,m^{-3}}$	Grazing half-saturation constant
L	$\mathrm{\mu mol\,m^{-2}\,s^{-1}}$	Local light intensity
L_{lim}	–	Light limitation factor
m_P	$\mathrm{d^{-1}}$	Phytoplankton mortality rate
m_Z	$(\mathrm{mmol\,N\,m^{-3}})^{-1}\,\mathrm{d^{-1}}$	Zooplankton mortality coefficient
n	–	Number of vertical layers
N	$\mathrm{mmol\,N\,m^{-3}}$	Dissolved inorganic nitrogen concentration in the water column
N_{lim}	–	Nutrient limitation factor

N_{n}	$\mathrm{mmol\,N\,m^{-3}}$	Dissolved inorganic nitrogen concentration in the bottom water cell
N_{sed}	$\mathrm{mmol\,N\,m^{-2}}$	Sediment nitrogen inventory
O	$\mathrm{mmol\,O_2\,m^{-3}}$	Dissolved oxygen concentration
O_{atm}	$\mathrm{mmol\,O_2\,m^{-3}}$	Atmospheric oxygen concentration
O_{crit}	$\mathrm{mmol\,O_2\,m^{-3}}$	Critical oxygen concentration
O_{lim}	–	Oxygen limitation factor
$O_{\mathrm{lim,n}}$	–	Oxygen limitation factor evaluated in the bottom water cell
P	$\mathrm{mmol\,N\,m^{-3}}$	Phytoplankton biomass concentration
$R(C)$	varies	Reaction term for tracer C
r_{ae}	$\mathrm{d^{-1}}$	Aerobic remineralization rate constant
r_{an}	$\mathrm{d^{-1}}$	Anaerobic remineralization rate constant
t	d	Time
w	$\mathrm{m\,d^{-1}}$	Generic vertical sinking velocity
w_D	$\mathrm{m\,d^{-1}}$	Detrital sinking velocity
y_P	$\mathrm{mmol\,O_2\,mmol\,N^{-1}}$	Stoichiometric oxygen yield associated with primary production
y_N	$\mathrm{mmol\,O_2\,mmol\,N^{-1}}$	Stoichiometric oxygen consumption associated with remineralization
z_i	m	Depth at the center of grid cell i
z_{sed}	m	Effective thickness of the sediment reservoir
Z	$\mathrm{mmol\,N\,m^{-3}}$	Zooplankton biomass concentration
ΔO	$\mathrm{mmol\,O_2\,m^{-3}}$	Transition width for oxygen limitation
Δz	m	Vertical grid spacing
ϵ_D	–	Fraction of grazing routed to detritus
ϵ_N	–	Fraction of grazing recycled to dissolved nutrients
κ	$\mathrm{m^2\,d^{-1}}$	Vertical diffusivity
μ	$\mathrm{d^{-1}}$	Realized phytoplankton growth rate
μ_{max}	$\mathrm{d^{-1}}$	Maximum phytoplankton growth rate

Usage of Large Language Models

AI was not used in this report for research purposes. ChatGPT was only used to rephrase already written abstracts. Since none of the authors are native English speakers, the tool helped improve spelling and wording to meet an academic standard.

Large Language Models were also used to produce part of the programming of the assignment.

Code Availability

The full model code used in this study is available on GitHub:

`https://github.com/matteo-mvh/Ecological-Modeling-Course`

Introduction

Marine ecosystem models are essential tools for understanding the interactions between biological production, nutrient cycling, and physical transport processes in the ocean. A commonly used simplified ecosystem representation is the NPZD framework, which includes nutrients (N), phytoplankton (P), zooplankton (Z), and detritus (D).

In this work the classical NPZD framework is extended with an oxygen tracer (O), resulting in an NPZDO model. Oxygen plays an important role in regulating remineralization and biological activity, particularly under low oxygen conditions.

The model is implemented as a one–dimensional vertical water column with seasonal forcing of light and mixing. Physical transport is represented by vertical diffusion and sinking of particulate material. The biological processes include phytoplankton growth, zooplankton grazing, mortality, detritus remineralization, and oxygen production and consumption.

The purpose of this report is to describe the structure of the NPZDO model, the governing equations, the numerical implementation, and the resulting ecosystem dynamics.

1 Model Description

1.1 Model Concept

The developed ecosystem model is a vertically resolved one-dimensional NPZDO model with an additional simplified representation of sediment biogeochemistry. The model describes the coupled dynamics of dissolved inorganic nitrogen, phytoplankton, zooplankton, detritus, dissolved oxygen, and benthic organic matter reservoirs.

The model is designed to capture the essential mechanisms controlling nutrient cycling, organic matter export, remineralization, and oxygen dynamics in a seasonally forced water column. In addition to pelagic ecosystem processes, the formulation explicitly represents the exchange of organic matter and nutrients between the water column and the sediments.

The water column is resolved on a vertical grid, whereas the sediment is represented by two vertically integrated reservoirs per unit area. This formulation allows the model to capture the first-order effects of benthic recycling without introducing additional vertical complexity.

1.1.1 State Variables

The model contains five prognostic state variables within the water column and two additional sediment reservoirs:

Symbol	State variable	Unit
N	Dissolved inorganic nitrogen	$\mathrm{mmol\,N\,m^{-3}}$
P	Phytoplankton biomass	$\mathrm{mmol\,N\,m^{-3}}$
Z	Zooplankton biomass	$\mathrm{mmol\,N\,m^{-3}}$
D	Detrital organic nitrogen	$\mathrm{mmol\,N\,m^{-3}}$
O	Dissolved oxygen	$\mathrm{mmol\,O_2\,m^{-3}}$
D_{sed}	Sediment detritus	$\mathrm{mmol\,N\,m^{-2}}$
N_{sed}	Sediment nutrients	$\mathrm{mmol\,N\,m^{-2}}$

All biological tracers except oxygen are expressed in nitrogen units to ensure consistent mass conservation between ecosystem compartments.

In addition to the water-column tracers, two sediment reservoirs are represented as vertically integrated pools per unit area and therefore do not possess a vertical profile.

1.1.2 Conceptual Illustration

Figure 1.1 presents a conceptual illustration of the NPZDO model structure and the main fluxes between compartments.

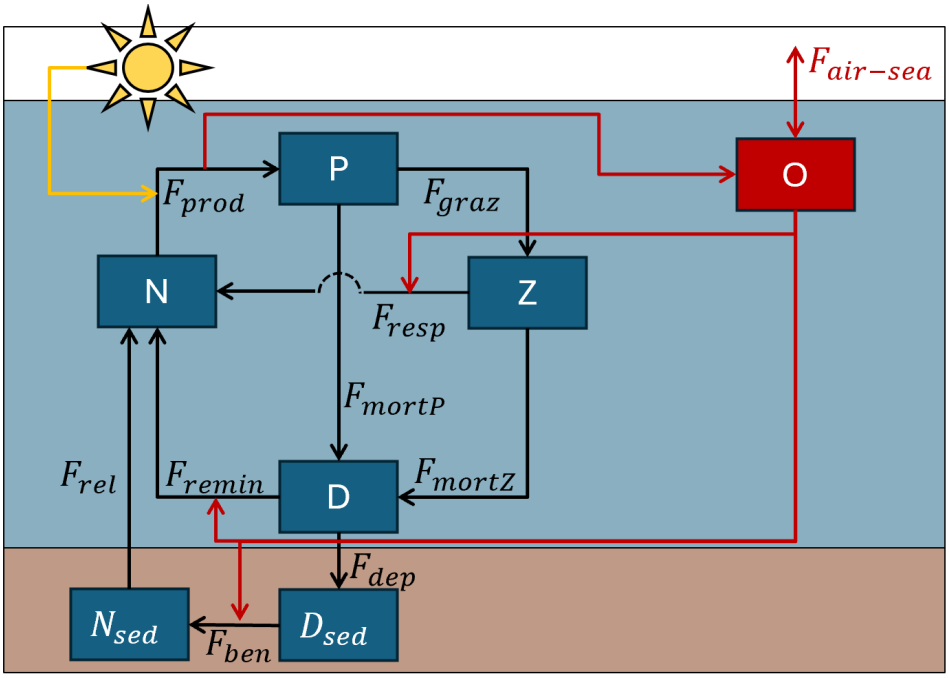

Figure 1.1: Schematic illustration of the NPZDO model showing the state variables and the biological, chemical, and sediment interactions represented in the model.

The schematic shown in Figure 1.1 summarizes the ecosystem compartments and fluxes represented in the NPZDO model. The pelagic ecosystem consists of dissolved inorganic nitrogen, phytoplankton, zooplankton, detritus, and dissolved oxygen. In addition, two sediment pools are included to represent the benthic nitrogen and detritus reservoirs, N_{sed} and D_{sed}.

Primary production converts dissolved inorganic nitrogen into phytoplankton biomass through the production flux F_{prod}, which is regulated by light availability and nutrient concentrations. Phytoplankton biomass is transferred to zooplankton through grazing, F_{graz}. A fraction of consumed phytoplankton is rapidly recycled to dissolved inorganic nitrogen, represented by the respiration and excretion flux F_{resp}.

Both phytoplankton and zooplankton contribute to the detrital pool through mortality processes, F_{mortP} and F_{mortZ}. The detritus pool represents sinking particulate organic matter in the water column. Detrital material is remineralized into dissolved inorganic

nitrogen through the remineralization flux F_{remin}, thereby recycling nutrients back into the pelagic nutrient pool.

Particulate detritus can also be exported from the water column to the sediments through the deposition flux F_{dep}. Once deposited, detritus enters the sediment detrital pool, where it undergoes benthic remineralization, F_{ben}. This process converts organic matter into dissolved inorganic nitrogen within the sediment nitrogen pool. A fraction of this benthic nitrogen is subsequently released back into the overlying water column via the release flux F_{rel}, providing an additional nutrient source to the pelagic nitrogen pool.

In addition to nitrogen cycling, the model explicitly represents oxygen dynamics. Oxygen is produced during photosynthesis associated with primary production and is consumed during respiration and remineralization processes. Furthermore, oxygen exchanges with the atmosphere through the air–sea gas exchange flux, $F_{\text{air-sea}}$.

Together, these processes represent the coupled dynamics between biological production, trophic interactions, organic matter export, sediment remineralization, and oxygen cycling within the modeled water column and sediment system.

1.1.3 Vertical Discretization

The water column in the model is represented by a one-dimensional vertical grid that extends from the ocean surface to the seabed. As illustrated in Figure 1.2, the depth axis is discretized into n vertically stacked computational cells. Each grid cell represents a finite depth interval centered at z_i, with neighboring cells connected through vertical transport processes such as diffusion and sinking.

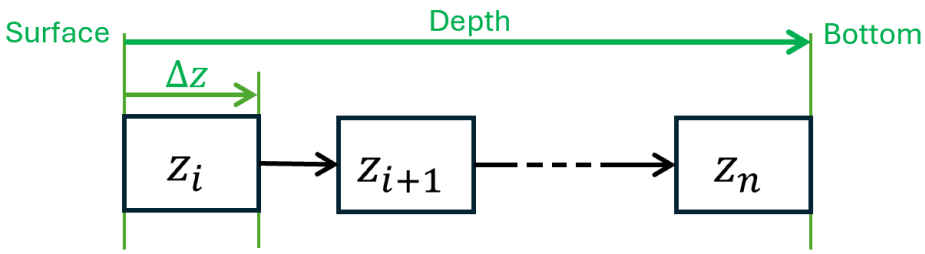

Figure 1.2: Schematic representation of the vertical discretization used in the model. The water column is divided into n depth cells extending from the surface to the bottom. Pelagic state variables (N, P, Z, D, and O) are defined in each grid cell, while sediment variables are represented as integrated reservoirs at the seabed.

The vertical grid is defined as

$$z_i = \left(i - \frac{1}{2}\right)\Delta z, \quad i = 1, 2, \ldots, n \tag{1.1}$$

where Δz is the grid spacing and n is the number of vertical layers.

All pelagic state variables in the model are resolved on this vertical grid. This includes dissolved inorganic nitrogen, phytoplankton biomass, zooplankton biomass, detritus, and dissolved oxygen. For each variable, concentrations are defined at the center of every grid cell, resulting in a vertical profile consisting of n concentrations that evolve dynamically over time.

Vertical transport between adjacent cells represents the exchange of material throughout the water column. This formulation allows the model to resolve vertical gradients in biological activity, nutrient availability, and oxygen concentration.

In contrast to the pelagic tracers, the sediment variables are not vertically resolved. The sediment nitrogen pool N_{sed} and sediment detritus pool D_{sed} are represented as vertically integrated quantities defined at the seabed. These variables are therefore treated as scalar reservoirs rather than depth-resolved profiles. Exchange between the water column and the sediment occurs through deposition and benthic remineralization fluxes applied at the bottom boundary of the vertical grid. While this simplified treatment is appropriate for the present model setup, a vertically resolved representation of sediment dynamics could provide a valuable extension in future work by allowing the model to capture internal sediment gradients and more detailed benthic processes.

1.2 Governing Equations

The biological dynamics of the NPZDO model can be expressed in terms of the fluxes illustrated in Figure 1.1. These fluxes represent the transfer of nitrogen between the ecosystem compartments and the exchange of organic matter with the sediment reservoir.

The local rate of change of the nitrogen-based state variables in the water column is given by

$$\frac{dN}{dt} = -F_{\text{prod}} + F_{\text{resp}} + F_{\text{remin}} + F_{\text{rel}}, \tag{1.2}$$

$$\frac{dP}{dt} = F_{\text{prod}} - F_{\text{graz}} - F_{\text{mortP}}, \tag{1.3}$$

$$\frac{dZ}{dt} = F_{\text{graz}} - F_{\text{mortZ}} - F_{\text{resp}}, \tag{1.4}$$

$$\frac{dD}{dt} = F_{\text{mortP}} + F_{\text{mortZ}} - F_{\text{remin}} - F_{\text{dep}}. \tag{1.5}$$

The temporal evolution of the sediment nitrogen and detritus reservoirs is described by

$$\frac{dD_{\text{sed}}}{dt} = F_{\text{dep}} - F_{\text{ben}}, \tag{1.6}$$

$$\frac{dN_{\text{sed}}}{dt} = F_{\text{ben}} - F_{\text{rel}}. \tag{1.7}$$

The sign convention of the individual flux terms follows directly from the conceptual illustration in Figure 1.1. Fluxes entering a compartment contribute positively to its rate of change, while fluxes leaving a compartment appear with a negative sign. In this way, the governing equations directly reflect the direction of the flux arrows shown in the schematic representation and ensure conservation of nitrogen between the modeled compartments.

Oxygen is included as an additional tracer in the model and is coupled to the biological processes through production during photosynthesis and consumption during respiration and remineralization. In addition, oxygen exchanges with the atmosphere through an air–sea flux at the surface boundary.

In addition to the biological transformations, all pelagic tracers are subject to vertical transport processes. The temporal evolution of each tracer is therefore described by a vertical advection–diffusion–reaction equation. In general form, the governing equation for a tracer concentration C may be written as

$$\frac{\partial C}{\partial t} = -\frac{\partial J}{\partial z} + R(C), \tag{1.8}$$

where J denotes the vertical tracer flux and $R(C)$ represents the reaction terms arising from biological processes. The transport term $-\partial J/\partial z$ accounts for vertical redistribution caused by turbulent diffusion and the sinking of particulate material, while the reaction term $R(C)$ describes the biological source and sink processes that transform nitrogen between the ecosystem compartments represented in the model.

1.2.1 Vertical Transport

The vertical flux of a tracer is formulated as the sum of an advective and a diffusive contribution:

$$J = wC - \kappa \frac{\partial C}{\partial z}, \tag{1.9}$$

where w is the sinking velocity and κ is the vertical diffusivity. Positive values of w correspond to downward transport. Substituting this flux definition into the general conservation equation gives

$$\frac{\partial C}{\partial t} = -\frac{\partial}{\partial z}\left(wC - \kappa \frac{\partial C}{\partial z}\right) + R(C). \tag{1.10}$$

In the present model, sinking is prescribed for phytoplankton and detritus, while nutrients, zooplankton, and oxygen do not experience vertical advection. Diffusivity varies seasonally and with depth, thereby representing changes in upper-ocean mixing intensity through the year.

1.2.2 Phytoplankton Growth

Primary production corresponds to the flux from dissolved nutrients to phytoplankton biomass and is denoted by F_{prod} in Figure 1.1. The production flux is defined as

$$F_{\text{prod}} = \mu P, \tag{1.11}$$

where μ is the realized phytoplankton growth rate. The growth rate is limited by both light availability and nutrient supply and is given by

$$\mu = \mu_{\max} \min(L_{\text{lim}}, N_{\text{lim}}), \tag{1.12}$$

where $\mu_{\max}$ is the maximum phytoplankton growth rate, L_{lim} is the light limitation factor, and N_{lim} is the nutrient limitation factor.

The limitation terms are expressed as Monod-type functions:

$$L_{\text{lim}} = \frac{L}{L + k_L}, \tag{1.13}$$

$$N_{\text{lim}} = \frac{N}{N + k_N}. \tag{1.14}$$

Here L represents the local light intensity and N the dissolved inorganic nitrogen concentration. The parameters k_L and k_N are half-saturation constants for light and nutrient uptake, respectively. Physically, these constants define the resource level at which phytoplankton growth reaches half of its maximum potential rate. Smaller values of k_L or k_N therefore indicate more efficient utilization of the corresponding resource.

This formulation implies that phytoplankton growth is limited by whichever of the two resources, light or nutrients, is most restrictive at a given depth and time.

1.2.3 Zooplankton Grazing

Zooplankton grazing represents the transfer of phytoplankton biomass to zooplankton and is denoted by the flux F_{graz} in Figure 1.1. The grazing flux is parameterized as

$$F_{\text{graz}} = g_{\max} \min(G_{\text{lim}}, O_{\text{lim}}) Z, \tag{1.15}$$

where $g_{\max}$ is the maximum grazing rate, G_{lim} is the prey limitation factor, and O_{lim} represents the oxygen limitation of grazing.

The prey limitation is given by

$$G_{\text{lim}} = \frac{P^2}{P^2 + k_Z^2}, \tag{1.16}$$

where P denotes the phytoplankton biomass and k_Z is the grazing half-saturation constant. Consequently, grazing increases with phytoplankton biomass but gradually saturates at high prey concentrations. The quadratic formulation introduces a sigmoidal response that suppresses grazing at low prey concentrations while allowing saturation at high phytoplankton biomass.

In addition to prey availability, grazing is also limited by oxygen availability. To avoid an abrupt threshold behaviour under low oxygen conditions, oxygen limitation is represented by a smooth hyperbolic tangent switch:

$$O_{\text{lim}} = \frac{1}{2}\left[1 + \tanh\left(\frac{O - O_{\text{crit}}}{\Delta O}\right)\right], \tag{1.17}$$

where O_{crit} is the critical oxygen concentration and ΔO determines the sharpness of the transition. This formulation allows grazing activity to gradually decline as oxygen concentrations decrease, representing reduced metabolic activity under hypoxic conditions.

The grazed phytoplankton biomass is not entirely converted into zooplankton biomass. Instead, the grazing flux is partitioned into three fractions to ensure mass conservation:

- a fraction ϵ_N is immediately recycled back into dissolved nutrients, representing rapid recycling through processes such as excretion and metabolic losses,
- a fraction ϵ_D is routed into the detrital pool, representing unassimilated organic matter produced during grazing,
- the remaining fraction $(1-\epsilon_N-\epsilon_D)$ contributes to zooplankton growth and represents the assimilated portion of the grazing flux.

This partitioning ensures that the full grazing flux is redistributed among the ecosystem

compartments while conserving nitrogen mass.

1.2.4 Mortality Fluxes

In addition to grazing, both phytoplankton and zooplankton experience mortality losses. These losses contribute to the detrital pool and are represented by the mortality fluxes

$$F_{\mathrm{mortP}} = m_P P, \tag{1.18}$$

$$F_{\mathrm{mortZ}} = m_Z Z^2. \tag{1.19}$$

Here, m_P is the phytoplankton mortality parameter and m_Z is the zooplankton mortality parameter. Phytoplankton mortality is modeled as a linear loss process, while zooplankton mortality follows a quadratic formulation. The quadratic dependence represents density-dependent losses such as predation or crowding effects that increase at higher zooplankton concentrations.

Both mortality fluxes transfer organic matter into the detritus pool, where it can subsequently be remineralized.

1.2.5 Water-Column Remineralization

Detrital organic matter in the water column is remineralized back into dissolved inorganic nitrogen through the combined remineralization flux F_{remin}. In the present model formulation, this flux consists of an aerobic and an anaerobic contribution acting simultaneously:

$$F_{\mathrm{remin}} = \underbrace{r_{\mathrm{ae}}\, O_{\mathrm{lim}}\, D}_{\text{aerobic}} + \underbrace{r_{\mathrm{an}}\, D}_{\text{anaerobic}} \,. \tag{1.20}$$

Here, D is the detritus concentration, r_{ae} and r_{an} are the aerobic and anaerobic remineralization rate constants, and O_{lim} is the oxygen limitation factor introduced in equation 1.17.

This formulation represents the simultaneous action of oxygen-dependent and oxygen-independent remineralization pathways. The aerobic contribution decreases under low-oxygen conditions through the factor O_{lim}, while the anaerobic contribution remains active independently of oxygen availability. As a result, total remineralization is reduced under oxygen-poor conditions, but does not vanish completely.

Only the aerobic component contributes directly to oxygen consumption. The corresponding oxygen sink in the water column is therefore linked to the aerobic fraction, while the

anaerobic fraction does not consume dissolved oxygen.

1.2.6 Benthic Remineralization

Once detrital material reaches the seabed, it enters the sediment detritus reservoir and is remineralized through the benthic remineralization flux F_{ben}. As in the water column, benthic remineralization is partitioned into aerobic and anaerobic components acting simultaneously:

$$F_{\mathrm{ben}} = \underbrace{r_{\mathrm{ae}}\, O_{\mathrm{lim,n}}\, D_{\mathrm{sed}}}_{\text{aerobic}} + \underbrace{r_{\mathrm{an}}\, D_{\mathrm{sed}}}_{\text{anaerobic}}. \tag{1.21}$$

Here, D_{sed} is the sediment detritus inventory and $O_{\mathrm{lim,n}}$ is the oxygen limitation evaluated in the bottom water. The total benthic remineralization flux therefore depends on both the amount of sedimented organic matter and the oxygen conditions at the sediment–water interface.

As for the water-column formulation, the aerobic benthic contribution is reduced under low-oxygen conditions, whereas the anaerobic contribution remains active. In the present model, only the aerobic benthic remineralization fraction contributes to oxygen consumption, which is applied as an oxygen sink in the bottom water cell.

The detritus supply to the sediment is described by the deposition flux

$$F_{\mathrm{dep}} = w_D D_{\mathrm{n}}, \tag{1.22}$$

where w_D is the detrital sinking velocity and D_{n} is the detritus concentration in the bottom water cell. This flux represents the downward transfer of particulate detritus from the water column into the sediment reservoir.

The return flux of dissolved inorganic nitrogen from the sediment to the bottom water is described by

$$F_{\mathrm{rel}} = -k_{N,\mathrm{n}} \left(\frac{N_{\mathrm{sed}}}{z_{\mathrm{sed}}} - N_{\mathrm{n}} \right), \tag{1.23}$$

where $k_{N,\mathrm{n}}$ is the sediment–water nitrogen exchange velocity, N_{sed} is the sediment nitrogen inventory, z_{sed} is the effective sediment thickness, and N_{n} is the dissolved inorganic nitrogen concentration in the bottom water cell. Following the model sign convention, positive fluxes are directed downward into the sediment. Accordingly, a negative value of F_{rel} corresponds to an upward release of dissolved nitrogen from the sediment back into

the water column.

1.2.7 Reaction Terms

Substituting the biological flux parameterizations introduced above into the nitrogen balance equations yields the internal source and sink terms acting within the water column.

The nutrient tendency in the water column is given by

$$\frac{dN}{dt} = \underbrace{-\mu P}_{\text{uptake}} + \underbrace{\epsilon_N g_{\max} \min(G_{\text{lim}}, O_{\text{lim}}) Z}_{\text{recycling}} + \underbrace{(r_{\text{ae}} O_{\text{lim}} D + r_{\text{an}} D)}_{\text{remineralization}}, \tag{1.24}$$

which represents nutrient uptake by phytoplankton, recycling of a fraction of the grazing flux, and remineralization of detrital organic matter within the water column.

The phytoplankton tendency is

$$\frac{dP}{dt} = \underbrace{\mu P}_{\text{production}} - \underbrace{g_{\max} \min(G_{\text{lim}}, O_{\text{lim}}) Z}_{\text{grazing}} - \underbrace{m_P P}_{\text{mortality}}, \tag{1.25}$$

where phytoplankton biomass increases through primary production and decreases through grazing and mortality.

The zooplankton tendency is

$$\frac{dZ}{dt} = \underbrace{(1 - \epsilon_N - \epsilon_D) g_{\max} \min(G_{\text{lim}}, O_{\text{lim}}) Z}_{\text{growth}} - \underbrace{m_Z Z^2}_{\text{mortality}}, \tag{1.26}$$

indicating that only the assimilated fraction of the grazing flux contributes to zooplankton growth, while quadratic mortality represents density-dependent losses.

The detritus tendency is

$$\frac{dD}{dt} = \underbrace{m_P P}_{\text{P mortality}} + \underbrace{\epsilon_D g_{\max} \min(G_{\text{lim}}, O_{\text{lim}}) Z}_{\text{egestion}} + \underbrace{m_Z Z^2}_{\text{Z mortality}} - \underbrace{(r_{\text{ae}} O_{\text{lim}} D + r_{\text{an}} D)}_{\text{remineralization}}, \tag{1.27}$$

where detritus is produced through phytoplankton mortality, zooplankton mortality, and the unassimilated fraction of grazing. Detritus is removed through remineralization processes within the water column.

1.3 Sediment Dynamics

The interaction between the water column and the seabed is represented through the exchange of organic matter and dissolved nutrients at the sediment–water interface. In the present model formulation, detrital organic matter reaching the bottom water cell is transferred into a vertically integrated sediment reservoir.

1.3.1 Sediment Boundary

The downward transport of detritus from the water column into the sediment is described by the deposition flux

$$F_{\mathrm{dep}} = w_D D_{\mathrm{n}}, \tag{1.28}$$

where w_D denotes the sinking velocity of detrital particles and D_{n} is the detritus concentration in the lowest water column grid cell. This formulation represents the settling of particulate organic matter that reaches the seabed and subsequently enters the sediment detritus pool.

In addition to organic matter deposition, dissolved inorganic nitrogen can be exchanged between the sediment reservoir and the bottom water. This exchange is parameterized through the flux

$$F_{\mathrm{rel}} = -k_{N,\mathrm{n}} \left(\frac{N_{\mathrm{sed}}}{z_{\mathrm{sed}}} - N_{\mathrm{n}} \right), \tag{1.29}$$

where $k_{N,\mathrm{n}}$ is the sediment–water exchange velocity, N_{sed} is the nitrogen inventory stored in the sediment reservoir, z_{sed} is the effective thickness of the sediment box, and N_{n} is the dissolved inorganic nitrogen concentration in the bottom water cell. The flux depends on the concentration difference between sediment porewater and bottom water, driving either a release of nutrients from the sediment or an uptake into the sediment.

1.3.2 Sediment Tendency

The temporal evolution of the sediment detritus pool results from the balance between deposition of organic matter from the water column and its subsequent remineralization within the sediment.

$$\frac{dD_{\text{sed}}}{dt} = \underbrace{w_D D_{\text{n}}}_{\text{deposition}} - \underbrace{(r_{\text{ae}} O_{\text{lim,n}} D_{\text{sed}} + r_{\text{an}} D_{\text{sed}})}_{\text{benthic remineralization}} . \tag{1.30}$$

The first term represents the supply of particulate organic matter to the sediment through gravitational settling of detritus from the water column. The second term represents the breakdown of organic matter within the sediment through benthic remineralization processes. As in the water column, remineralization is partitioned into an aerobic component that depends on oxygen availability and an anaerobic component that proceeds independently of oxygen.

The sediment nitrogen reservoir evolves according to

$$\frac{dN_{\text{sed}}}{dt} = \underbrace{(r_{\text{ae}} O_{\text{lim,n}} D_{\text{sed}} + r_{\text{an}} D_{\text{sed}})}_{\text{remineralization}} - \underbrace{k_{N,\text{n}} \left(\frac{N_{\text{sed}}}{z_{\text{sed}}} - N_{\text{n}} \right)}_{\text{exchange}} . \tag{1.31}$$

The first term represents the production of dissolved inorganic nitrogen through the remineralization of sediment detritus. The second term describes the exchange of this dissolved nitrogen between the sediment reservoir and the overlying bottom water, which may either release nutrients back to the water column or transfer nitrogen into the sediment depending on the concentration gradient across the sediment–water interface.

1.4 Oxygen Dynamics

In addition to the nitrogen-based ecosystem tracers, the model explicitly represents dissolved oxygen in the water column. Oxygen dynamics are controlled by biological production and consumption processes as well as exchange with the atmosphere at the sea surface.

1.4.1 Oxygen Exchange at the Boundaries

At the air–sea interface, oxygen exchanges with the atmosphere through a restoring-type gas exchange flux,

$$F_{\text{air-sea}} = k_{\text{surf}} \left(O_{\text{atm}} - O_{\text{surf}} \right) , \tag{1.32}$$

where k_{surf} is the surface gas transfer coefficient, O_{atm} is the equilibrium oxygen concentration imposed by the atmosphere, and O_{surf} is the oxygen concentration in the uppermost model cell. When surface waters are undersaturated relative to the atmosphere, this

formulation leads to a flux of oxygen into the ocean, whereas supersaturated conditions result in oxygen outgassing.

At the seabed, oxygen is consumed through aerobic remineralization of organic matter within the sediment. Detritus that reaches the bottom water cell is transferred into the sediment reservoir and subsequently remineralized through the benthic remineralization flux. The aerobic component of this process consumes oxygen in the overlying bottom water and therefore imposes an oxygen demand at the sediment–water interface.

The resulting bottom oxygen flux can be written as

$$F_{\text{sed}} = -y_N r_{\text{ae}} O_{\text{lim,n}} D_{\text{sed}}, \tag{1.33}$$

where the negative sign indicates a flux of oxygen from the bottom water into the sediment due to aerobic remineralization. This formulation links the benthic oxygen consumption directly to the aerobic remineralization rate of the sediment detritus pool D_{sed}.

1.4.2 Oxygen Tendency Equation

Oxygen production and consumption are linked to the nitrogen-based biological fluxes through stoichiometric coefficients. The parameter y_P represents the amount of oxygen produced per unit of nitrogen incorporated into phytoplankton biomass during primary production, while y_N defines the oxygen consumption associated with respiration and aerobic remineralization of organic nitrogen.

These coefficients ensure a consistent coupling between the nitrogen and oxygen cycles within the model. The oxygen tendency associated with internal biological processes in the water column can be written as

$$\frac{dO}{dt} = \underbrace{y_P \mu P}_{\text{production}} - \underbrace{y_N \epsilon_N g_{\max} \min(G_{\text{lim}}, O_{\text{lim}}) Z}_{\text{respiration}} - \underbrace{y_N r_{\text{ae}} O_{\text{lim}} D}_{\text{remineralization}}. \tag{1.34}$$

The first term represents oxygen production during phytoplankton growth. The second term describes oxygen consumption associated with respiration and rapid recycling of grazed phytoplankton biomass. The third term represents oxygen consumption during aerobic remineralization of detritus within the water column.

Additional oxygen sources and sinks arise from boundary fluxes at the air–sea interface and at the sediment–water interface. These contributions are applied separately through the boundary conditions described in Section 1.4.1.

2 Model Setup and Convergence

2.1 Numerical Implementation

The NPZDO ecosystem model is solved on a one-dimensional vertical grid extending over different water depths. The water column is discretized into n_z equally spaced grid cells, and all pelagic state variables (N, P, Z, D, and O) are defined at the cell centres.

Vertical transport is formulated in flux form using the tracer flux, as introduced in Equation 1.9, which ensures a conservative discretization of both turbulent diffusion and sinking transport. The governing equations are therefore discretized in a finite-volume-like formulation where the divergence of the flux determines the change in tracer concentration within each grid cell.

After spatial discretization, the model equations form a system of coupled ordinary differential equations of the form

$$\frac{d\mathbf{y}}{dt} = f(\mathbf{y}, t), \tag{2.1}$$

where the state vector $\mathbf{y}$ contains the vertical profiles of all prognostic tracers as well as the sediment reservoirs D_{sed} and N_{sed}.

Time integration is carried out using the implicit backward differentiation formula (BDF) solver available in `SciPy`. This solver is well suited for the present model, since the system combines relatively fast biological reactions with diffusive transport and nonlinear source terms, which together lead to stiff dynamics.

2.1.1 Boundary Conditions

At the air–sea interface, oxygen is allowed to exchange with the atmosphere through a restoring-type air–sea flux,

$$J_{\text{surf}} = k_{\text{surf}} \left(O_{\text{atm}} - O_{\text{surf}} \right), \tag{2.2}$$

where k_{surf} is the surface gas exchange coefficient, O_{atm} is the prescribed atmospheric equilibrium oxygen concentration, and O_{surf} is the oxygen concentration in the uppermost model cell.

At the seabed, detritus reaching the bottom grid cell is transferred into the sediment reservoir through the deposition flux

$$F_{\mathrm{dep}} = w_D D_{\mathrm{n}}, \tag{2.3}$$

where D_{n} is the detritus concentration in the bottom water cell. Within the sediment reservoir, detrital material is remineralized and converted into dissolved inorganic nitrogen, which can subsequently be released back into the bottom water via the exchange flux

$$F_{\mathrm{rel}} = -k_{N,\mathrm{n}} \left(\frac{N_{\mathrm{sed}}}{z_{\mathrm{sed}}} - N_{\mathrm{n}} \right). \tag{2.4}$$

In addition, benthic remineralization imposes an oxygen demand on the bottom water cell through the aerobic component of the sediment remineralization flux.

For the remaining tracers (N, P, Z, and D), zero-gradient boundary conditions are applied at the seabed for diffusive transport, ensuring that no artificial diffusive flux occurs across the bottom boundary.

2.2 Convergence Checks

To verify that the numerical solution is robust and independent of discretization choices, several convergence tests were performed. These tests include a grid convergence analysis as well as two complementary time convergence diagnostics.

2.2.1 Grid Convergence

To assess the sensitivity of the solution to vertical resolution, the model was integrated using different numbers of vertical layers while keeping all model parameters, forcing functions, and solver settings identical. The tested grid sizes ranged from $n_z = 10$ to $n_z = 60$ layers over the 200 m water column.

For each simulation, the resulting tracer fields were compared against the highest-resolution simulation ($n_z = 60$), which was used as the reference solution. Differences were quantified using the relative L_2 error norm. For a tracer field C, the relative error between a simulation and the reference solution is defined as

$$E_{L_2} = \frac{\sqrt{\sum_{i=1}^{n_z} \left(C_i - C_i^{\text{ref}}\right)^2}}{\sqrt{\sum_{i=1}^{n_z} \left(C_i^{\text{ref}}\right)^2}}, \tag{2.5}$$

where C_i denotes the tracer value in grid cell i for the tested simulation and C_i^{ref} is the corresponding value obtained from the reference simulation with the highest vertical resolution. This formulation measures the deviation of the numerical solution relative to the magnitude of the reference field. The diagnostic was evaluated for several quantities, including the final vertical profiles of phytoplankton and oxygen as well as the bottom oxygen time series during the final simulation year.

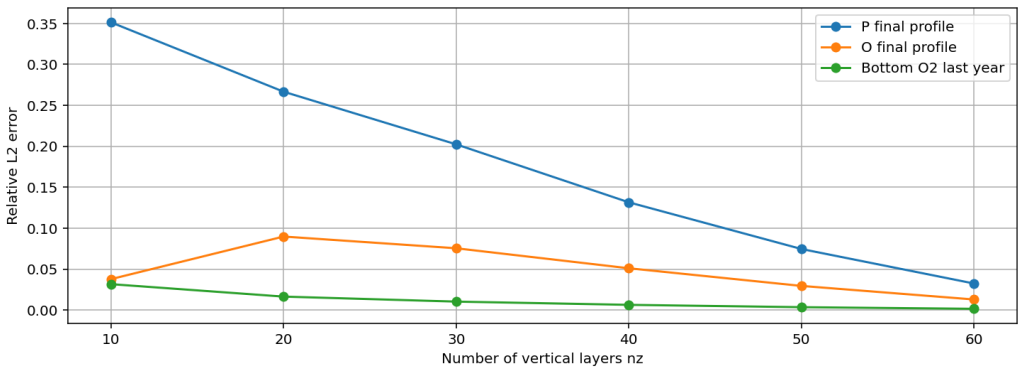

Figure 2.1: Grid convergence analysis showing the relative L_2 error of selected diagnostics compared to the highest-resolution simulation.

As shown in Figure 2.1, the relative error decreases systematically as the vertical resolution increases, indicating that the numerical solution approaches a grid-independent limit. Among the evaluated quantities, the phytoplankton profile shows the strongest sensitivity to grid resolution, while oxygen-related diagnostics converge more rapidly.

Based on this analysis, a resolution of $n_z = 50$ layers was selected for the remaining simulations. At this resolution the phytoplankton error falls below 0.1 relative to the reference solution, while oxygen-related quantities already show very small deviations. This choice therefore provides results close to the converged solution while maintaining reasonable computational efficiency.

2.2.2 Time Convergence

In addition to spatial convergence, it is important to ensure that the system approaches a statistically steady seasonal cycle. To evaluate this behaviour, the vertically integrated nitrogen pools were compared between successive simulation years.

For each year i, the depth-integrated inventories of N, P, Z, and D were computed and the differences between consecutive years were evaluated. The total nitrogen inventory,

$$N_{\text{tot}} = \int (N + P + Z + D)\, dz, \tag{2.6}$$

was also monitored as a diagnostic of nitrogen conservation.

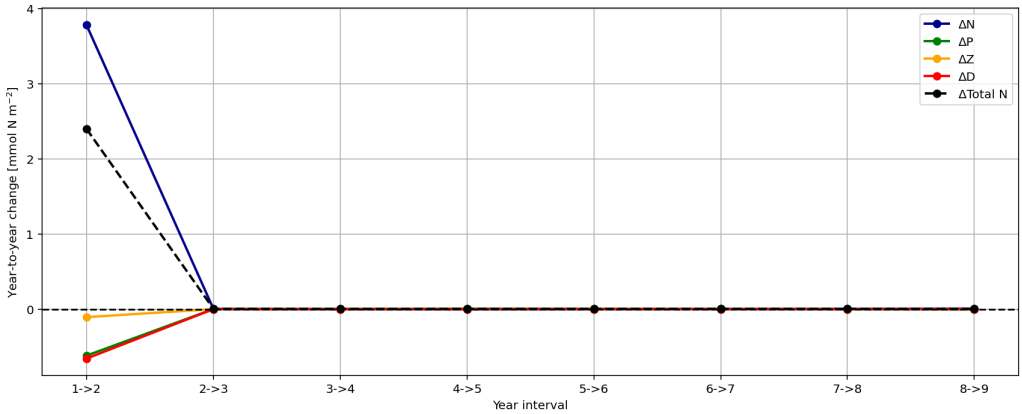

Figure 2.2: Year-to-year changes in vertically integrated nitrogen pools during the model spin-up.

Figure 2.2 shows that the year-to-year differences decrease rapidly during the initial years of the simulation and become negligible afterwards. After approximately two years of integration, all nitrogen pools fluctuate only within machine precision, indicating that the system has reached a stable seasonal cycle. A visual inspection of the time series further confirms that the seasonal dynamics remain consistent between successive years, with no noticeable drift in the system state. Based on these results, a total simulation length of four years was chosen for the subsequent experiments, consisting of three years of model spin-up followed by one additional year used for analysis.

3 Model Behaviour and Analysis

3.1 Factors Limiting Phytoplankton Growth

As introduced in Sections 1.2.2 and 1.2.3, the biological fluxes and state variables in the model are regulated by several environmental limitation functions. These limitations determine the realized growth and interaction rates of the ecosystem components and therefore play a central role in shaping the simulated ecosystem dynamics.

Within the present model formulation, phytoplankton growth is primarily controlled by two key resources: light availability and nutrient supply. The realized phytoplankton growth rate is determined by the minimum of the light and nutrient limitation terms. This formulation reflects the classical ecological principle known as *Liebig's law of the minimum*, which states that biological growth is limited by the scarcest resource rather than the total availability of all resources. Consequently, rapid phytoplankton growth can only occur when both light and nutrients are sufficiently available.

During winter, deep mixing and weak surface irradiance strongly reduce the light available within the mixed layer. Under these conditions, light limitation becomes the dominant control on phytoplankton growth. In contrast, during and after the spring bloom, nutrients in the upper water column are progressively depleted by biological uptake. Nutrient limitation therefore becomes increasingly important, particularly during summer when vertical stratification suppresses the resupply of nutrients from deeper waters.

In addition to these bottom-up controls, other processes indirectly influence phytoplankton dynamics. Grazing by zooplankton removes phytoplankton biomass and represents an important *top-down control* on primary producers. The balance between nutrient supply and grazing pressure therefore determines the overall ecosystem state and reflects the classical ecological distinction between *bottom-up versus top-down control* in marine food webs.

As phytoplankton biomass increases during bloom conditions, the organisms themselves contribute to the attenuation of light within the water column. This process, often referred to as *self shading*, reduces the penetration of light into deeper layers and can suppress growth below the surface. Self shading therefore acts as an important negative feedback that limits further biomass accumulation during intense blooms.

Furthermore, organic matter produced during primary production is transferred into the detrital pool through mortality and grazing losses. The subsequent remineralization of this detritus releases dissolved nutrients back into the water column, forming an important *nutrient recycling* pathway that partially sustains phytoplankton production even when external nutrient inputs are limited.

Together, the interaction between light availability, nutrient supply, grazing pressure, self shading, and nutrient recycling determines the overall behaviour of the NPZDO ecosystem model and ultimately controls the vertical structure and seasonal evolution of biomass and nutrient concentrations.

3.1.1 Limitation Functions

The limitation functions used in the model are shown in Figure 3.1. Different functional forms were chosen to represent the underlying biological and biochemical processes.

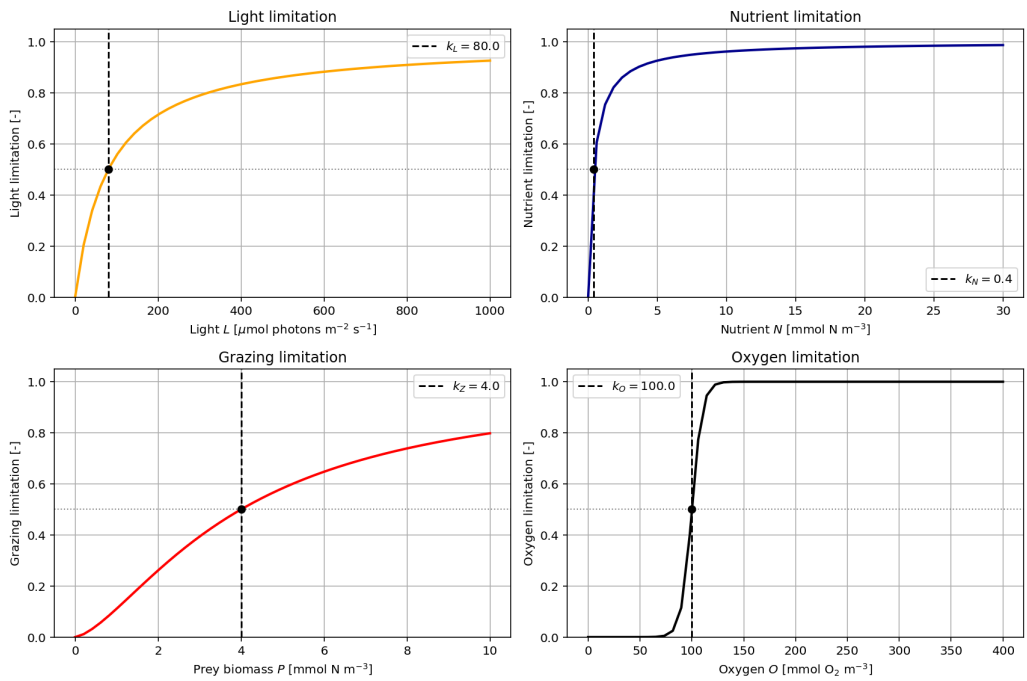

Figure 3.1: Functional forms of the limitation terms used in the ecosystem model.

Light and nutrient limitation are represented using classical saturation functions (Type II functional responses). These functions are widely used in ecosystem models because they capture the biochemical nature of resource uptake, where growth increases with resource availability but gradually approaches a maximum rate.

Oxygen limitation is represented using the hyperbolic tangent formulation introduced in Equation 1.17. This formulation introduces a critical oxygen threshold at which metabolic processes rapidly decline. As oxygen concentrations fall below this threshold, zooplankton

activity and aerobic remineralization are strongly reduced.

The grazing limitation follows a Type III functional response, which exhibits a slower increase at low prey concentrations. This formulation reflects the ecological difficulty for zooplankton to locate and capture prey when phytoplankton biomass is low, while allowing grazing to increase rapidly once prey becomes abundant. Such nonlinear grazing behaviour is often important for regulating bloom dynamics, as grazing pressure increases strongly once phytoplankton concentrations rise.

3.1.2 Vertical Structure of Limitation

To illustrate how these limitation mechanisms interact in the model, Figure 3.2 shows a representative vertical profile of the limitation terms and phytoplankton biomass during a typical summer situation.

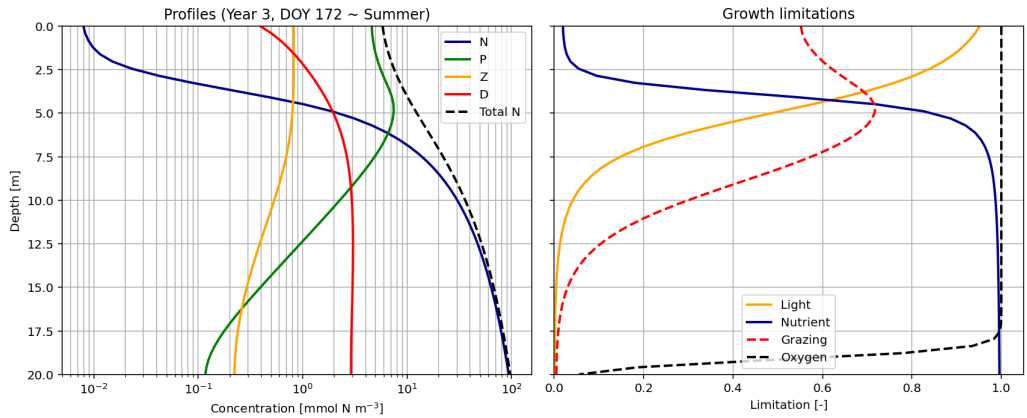

Figure 3.2: Vertical distribution of limitation factors and phytoplankton biomass for a representative summer state of the model.

The figure illustrates the characteristic vertical structure of phytoplankton limitation. Phytoplankton biomass typically reaches its maximum at depths where light and nutrient availability are both sufficiently high. In the model, this corresponds to the depth where the light and nutrient limitation functions intersect, resulting in optimal growth conditions.

The influence of grazing limitation is less directly visible in this profile, as zooplankton dynamics depend primarily on prey availability rather than environmental resources. However, grazing plays an important regulatory role by preventing excessive accumulation of phytoplankton biomass and by transferring organic matter to higher trophic levels and the detrital pool.

Oxygen limitation becomes important under strongly hypoxic conditions. At such depths, low oxygen concentrations can suppress zooplankton activity and reduce aerobic reminer-

alization rates. This in turn influences the rate at which organic matter is recycled back into dissolved nutrients.

3.2 Ecosystem Response to Nutrient Loading

To investigate how external nutrient supply influences the structure of the ecosystem, the model was integrated under constant summer forcing until a steady-state solution was reached. Two simulations were performed representing contrasting nutrient loading scenarios: a low nutrient input case and a high nutrient input case. The resulting vertical distributions of nutrients, oxygen, phytoplankton, and detritus are shown in Figure 3.3.

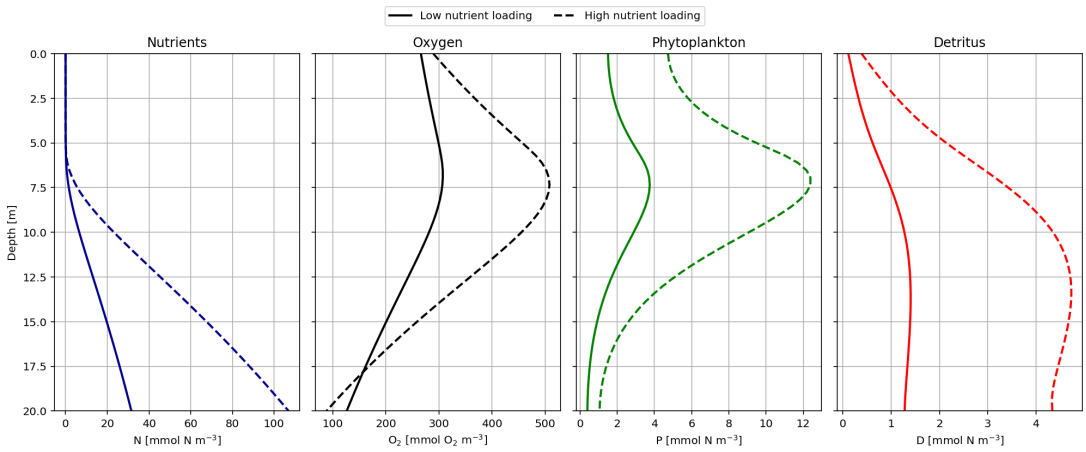

Figure 3.3: Steady-state vertical profiles of nutrients, oxygen, phytoplankton, and detritus for low (solid lines) and high (dashed lines) nutrient loading under constant summer forcing.

The simulations reveal clear differences in the vertical structure of the ecosystem between the two nutrient regimes. Under high nutrient loading, nutrient concentrations remain elevated throughout the water column compared to the low-nutrient case. As a consequence, phytoplankton growth is less strongly limited by nutrient availability and substantially higher phytoplankton biomass develops in the upper water column.

From an ecological perspective, this situation represents a system dominated by strong bottom-up control, where the availability of nutrients directly regulates the magnitude of primary production. However, even under high nutrient availability phytoplankton biomass does not increase indefinitely. Instead, several regulatory mechanisms emerge that stabilize the ecosystem.

One important process is grazing control. As phytoplankton biomass increases, zooplankton populations respond to the elevated food availability and grazing pressure intensifies. This top-down control limits the accumulation of phytoplankton biomass and transfers organic matter into higher trophic levels and the detrital pool.

A second stabilizing mechanism is self shading. Increasing phytoplankton concentrations enhance light attenuation within the water column, which reduces light availability in deeper layers and limits further growth. This density-dependent feedback prevents unlimited biomass accumulation even when nutrients remain abundant.

The increased primary production in the high-nutrient case also leads to a corresponding increase in detritus production through mortality and grazing processes. As detrital particles sink through the water column, they are remineralized by microbial processes. This enhanced remineralization represents an important nutrient recycling pathway that returns dissolved nutrients to the water column.

At the same time, remineralization consumes dissolved oxygen below the productive layer. Consequently, oxygen concentrations decrease more strongly with depth in the high-nutrient simulation compared to the low-nutrient case.

In contrast, the low nutrient loading scenario produces a much weaker phytoplankton bloom. Nutrient limitation restricts primary production in the upper water column, resulting in lower phytoplankton biomass and reduced detritus formation. With less organic material available for remineralization, oxygen consumption in the deeper water column remains comparatively weak and oxygen concentrations stay higher throughout the water column.

Overall, the differences between the two steady-state solutions arise from the coupling between nutrient availability, primary production, organic matter export, and oxygen consumption. Higher nutrient loading stimulates primary production, which increases the flux of organic matter through the ecosystem and ultimately enhances oxygen consumption through remineralization processes at depth.

3.3 Seasonal Ecosystem Dynamics

When the model is forced with seasonally varying environmental conditions, the ecosystem evolves through a characteristic annual cycle. The seasonal forcing functions used in the model are illustrated in Figure 3.4. These forcing terms prescribe the seasonal variation of vertical diffusivity, surface light intensity, and air–sea gas exchange. Together they define the physical environment in which the biological ecosystem operates and strongly influence the vertical transport of nutrients, organic matter, and oxygen.

A key driver of the seasonal dynamics is the variation in vertical diffusivity κ, which controls the strength of turbulent mixing and therefore the degree of stratification in the water column. During winter, high diffusivity values represent strong mixing conditions that homogenize the upper water column and allow nutrients from deeper layers to be transported toward the surface. In contrast, diffusivity is significantly reduced

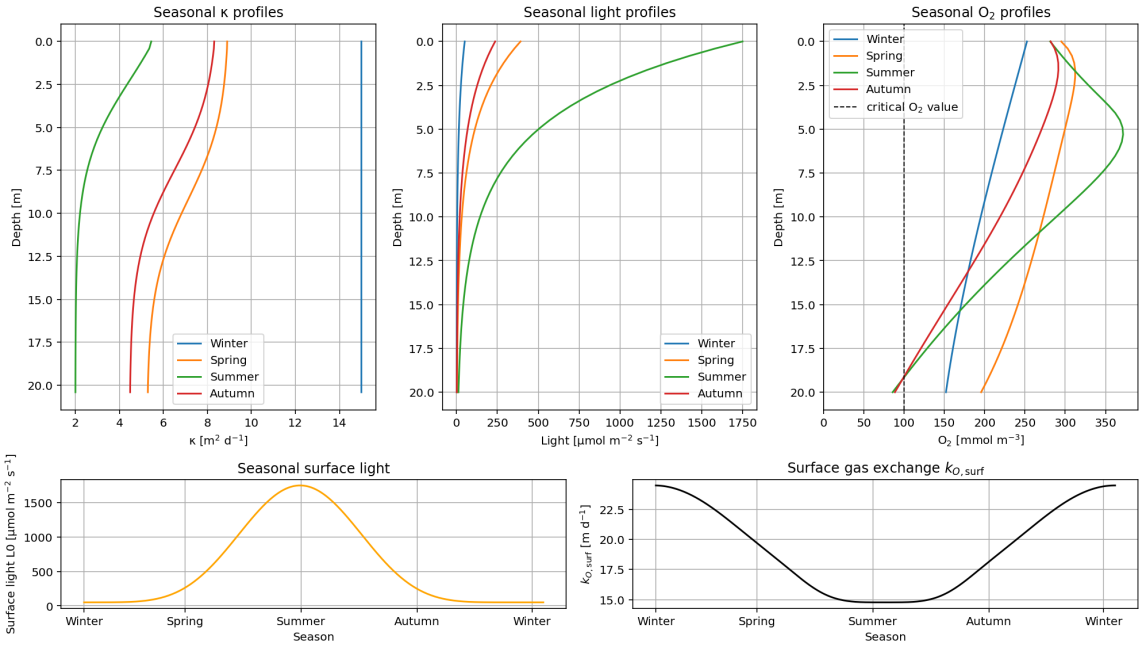

Figure 3.4: Seasonal forcing used in the model. Shown are the seasonal vertical diffusivity profiles, seasonal light attenuation profiles, the annual cycle of surface light intensity, and the seasonal variation of the surface gas exchange coefficient. The oxygen profiles represent the resulting seasonal oxygen distribution produced by the model rather than an externally prescribed forcing.

during summer, leading to a more stratified water column in which vertical transport is suppressed. This seasonal change in mixing intensity strongly influences the vertical distribution of nutrients and biological tracers.

At the same time, the seasonal cycle of surface light intensity controls the availability of energy for primary production. Surface irradiance increases from winter toward summer, allowing light to penetrate deeper into the water column and expanding the depth range in which phytoplankton growth can occur. Light availability therefore determines when and where primary production becomes possible, while vertical mixing regulates the resupply of nutrients required to sustain that production.

The oxygen field shown in Figure 3.4 is not imposed as an external forcing but represents the simulated seasonal oxygen distribution produced by the model. Oxygen concentrations evolve dynamically as a result of the interaction between biological production during photosynthesis, oxygen consumption through respiration and remineralization, vertical mixing, and exchange with the atmosphere at the sea surface.

The resulting seasonal evolution of the ecosystem state variables is shown in Figure 3.5. The panels illustrate the temporal and vertical evolution of dissolved inorganic nitrogen, phytoplankton, zooplankton, detritus, and oxygen over the annual cycle.

During winter, low surface light levels strongly limit phytoplankton growth throughout

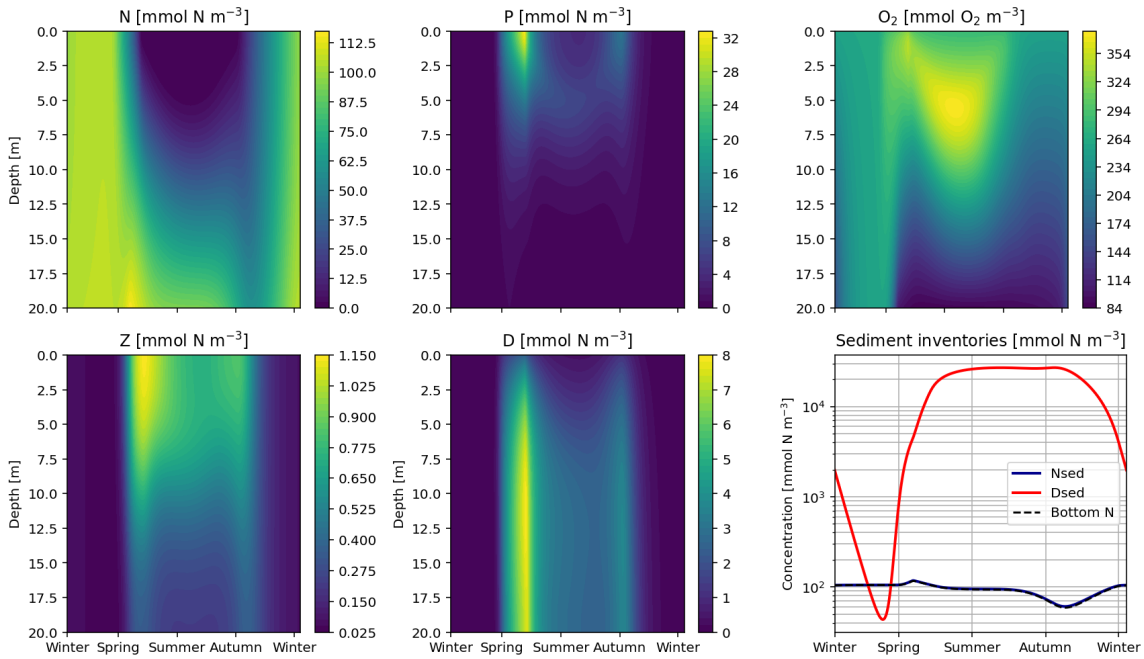

Figure 3.5: Seasonal evolution of ecosystem state variables. The panels show the time–depth distribution of nutrients, phytoplankton, oxygen, zooplankton, detritus, and sediment inventories over one annual cycle. The simulation represents strongly eutrophic conditions with elevated nutrient availability, which were intentionally chosen in order to investigate the development of oxygen depletion and anoxic conditions within the water column and near the seabed.

the water column. As a consequence, phytoplankton biomass remains very low and nutrient concentrations remain high. Strong vertical mixing distributes nutrients relatively uniformly throughout the water column, preventing the formation of strong vertical gradients. Zooplankton biomass and detritus concentrations are also minimal during this period because primary production is weak.

In early spring, increasing surface irradiance rapidly improves growth conditions in the upper ocean. Once light limitation weakens, phytoplankton growth accelerates and a pronounced spring bloom develops near the surface. This bloom is clearly visible in Figure 3.5 as a sharp increase in phytoplankton biomass within the upper water column. As phytoplankton biomass increases, zooplankton grazing intensifies and detritus production rises due to mortality and unassimilated grazing.

During summer, the ecosystem transitions into a nutrient-limited regime. The strong spring bloom depletes nutrients in the surface layer, while reduced vertical mixing limits the resupply of nutrients from deeper waters. As a result, phytoplankton biomass decreases and the productive layer becomes confined to shallow depths where light remains sufficient. At the same time, the detrital material produced during the bloom sinks downward and undergoes remineralization. This process consumes oxygen below the productive layer and leads to a gradual reduction of oxygen concentrations at intermediate

depths.

Under the strongly stratified summer conditions, vertical mixing is weak and the transport of oxygen from the surface to deeper layers becomes limited. At the same time, the elevated detritus flux generated during the spring bloom fuels intensified aerobic remineralization below the euphotic zone. This combination of reduced oxygen supply and enhanced biological oxygen consumption results in pronounced oxygen depletion at depth. In the present eutrophic simulation, oxygen concentrations in the bottom layers can approach hypoxic or even anoxic conditions, illustrating how strong primary production and subsequent organic matter remineralization can drive seasonal oxygen deficits in stratified coastal systems.

The seasonal structure of the ecosystem can also be seen in the representative vertical profiles shown in Figure 3.6. These profiles illustrate the typical vertical distribution of phytoplankton, zooplankton, and detritus during the four seasons.

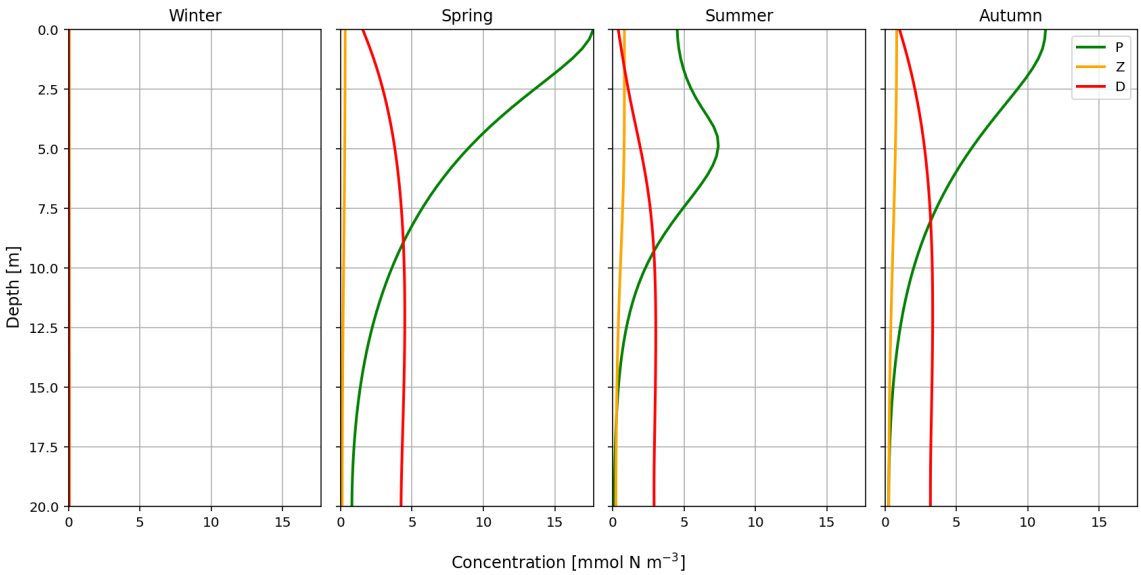

Figure 3.6: Representative vertical profiles of phytoplankton, zooplankton, and detritus during winter, spring, summer, and autumn.

The winter profiles show almost negligible biomass throughout the water column due to strong light limitation. In spring, phytoplankton biomass increases rapidly in the upper layers, forming the characteristic bloom structure. Zooplankton populations subsequently increase in response to the elevated food supply, while detritus concentrations rise due to increased mortality and grazing losses.

During summer, the phytoplankton maximum becomes weaker and shifts slightly deeper as surface nutrients become depleted. Zooplankton biomass remains relatively low because prey concentrations decline after the bloom peak. Detritus, however, continues to accumulate below the productive layer due to the sinking flux of organic matter.

In autumn, increasing vertical mixing begins to restore nutrients in the upper water column. Although light levels are already decreasing, this partial nutrient resupply can lead to a modest secondary increase in phytoplankton biomass before winter conditions return.

Overall, the simulated seasonal cycle follows the classical temperate marine ecosystem pattern: winter nutrient accumulation, a strong spring phytoplankton bloom, summer nutrient limitation with continued remineralization at depth, and partial ecosystem recovery during autumn mixing.

3.4 Sensitivity Analysis

In order to assess how strongly the simulated ecosystem dynamics depend on the chosen parameter values, a series of sensitivity experiments was performed. In these experiments, selected biological parameters were varied while keeping all other model parameters and forcing conditions identical. The resulting steady-state vertical profiles of nutrients, phytoplankton, oxygen, and detritus were then compared to the reference simulation.

Sensitivity analyses are an important component of ecosystem modelling because many biological parameters are only approximately known and can vary between ecosystems. Exploring how model solutions respond to parameter changes helps identify which processes exert the strongest control on ecosystem structure and biogeochemical cycling.

In the present model, three parameters were investigated in particular: phytoplankton mortality, the nutrient half-saturation constant for phytoplankton uptake, and the sinking velocity of detrital particles. Each of these parameters represents a different ecological mechanism influencing the balance between production, recycling, and export of organic matter.

3.4.1 Sensitivity to Phytoplankton Mortality

The first experiment examines the sensitivity of the system to the phytoplankton mortality rate m_P. Figure 3.7 compares steady-state vertical profiles for two different mortality rates.

Phytoplankton mortality represents a direct loss pathway from the living biomass pool and transfers organic matter into the detrital pool. Mechanistically, this pathway can be summarized as

$$P \xrightarrow{m_P} D \xrightarrow{r} N, \tag{3.1}$$

where phytoplankton biomass P is converted into detritus D through mortality and detritus is subsequently remineralized back into dissolved nutrients N at rate r. Increasing the mortality rate therefore accelerates the transfer of organic matter from the living biomass pool to detritus and strengthens the recycling pathway within the ecosystem.

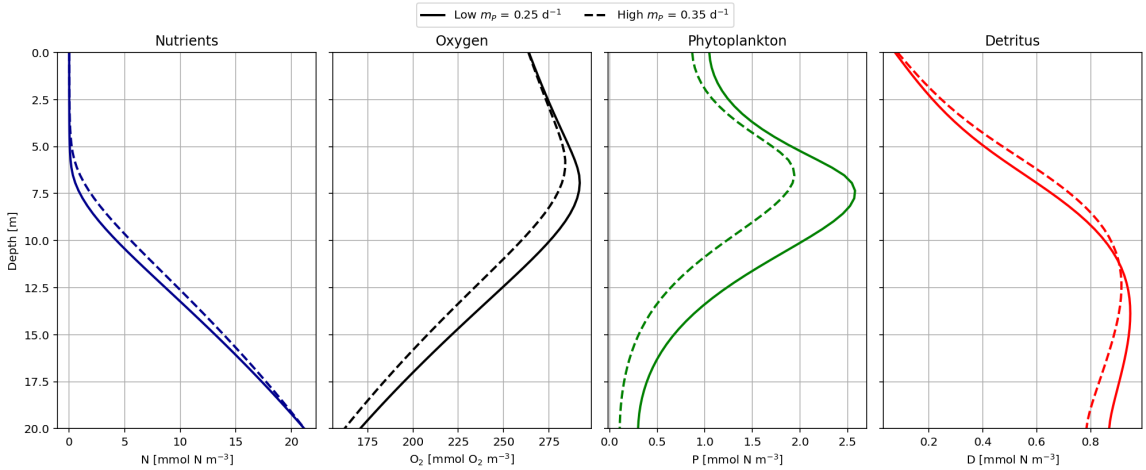

Figure 3.7: Sensitivity of the ecosystem structure to changes in phytoplankton mortality rate.

As phytoplankton mortality increases, the standing stock of phytoplankton decreases throughout the water column because biomass is removed more rapidly. At the same time, the enhanced transfer of organic material into the detrital pool increases the rate at which nutrients are returned to the dissolved pool through remineralization. This explains why nutrient concentrations remain slightly higher when mortality rates are increased.

An additional effect of higher mortality is a slight upward shift of the phytoplankton maximum toward shallower depths. Because biomass losses increase everywhere in the water column, phytoplankton populations can only persist where growth conditions are strongest, which typically occurs closer to the surface where light availability is highest. As a consequence, the balance between growth and mortality shifts upward in the water column.

The enhanced detritus production also increases the amount of organic matter that undergoes remineralization below the productive layer. Since aerobic remineralization consumes oxygen, this leads to slightly stronger oxygen depletion at intermediate depths.

Overall, this experiment highlights the important role of phytoplankton loss processes in regulating both the magnitude and vertical distribution of biomass, while simultaneously influencing the strength of nutrient recycling and oxygen consumption within the ecosystem.

3.4.2 Sensitivity to Nutrient Uptake Efficiency

The second experiment investigates the influence of the nutrient half-saturation constant k_N, which controls how efficiently phytoplankton can utilize dissolved nutrients at low concentrations. The influence of this parameter can be understood directly from the phytoplankton growth formulation

$$\mu = \mu_{\max} \min\left(\frac{L}{L + k_L}, \frac{N}{N + k_N}\right), \tag{3.2}$$

where k_N determines how rapidly the nutrient limitation term approaches saturation. Larger values of k_N reduce the nutrient affinity of phytoplankton, meaning that higher nutrient concentrations are required to sustain the same growth rate.

Increasing k_N therefore strengthens nutrient limitation and slightly reduces phytoplankton growth rates throughout the water column. Mechanistically, the effect can be summarized as

$$k_N \uparrow \Rightarrow \text{ nutrient limitation} \uparrow \Rightarrow \; P \downarrow \Rightarrow \text{ nutrient uptake} \downarrow . \tag{3.3}$$

As phytoplankton biomass decreases, the biological uptake of nutrients is reduced, allowing dissolved nutrients to remain at somewhat higher concentrations. Because the overall magnitude of primary production changes only moderately, the resulting detritus concentrations and oxygen distributions remain largely similar between the two simulations.

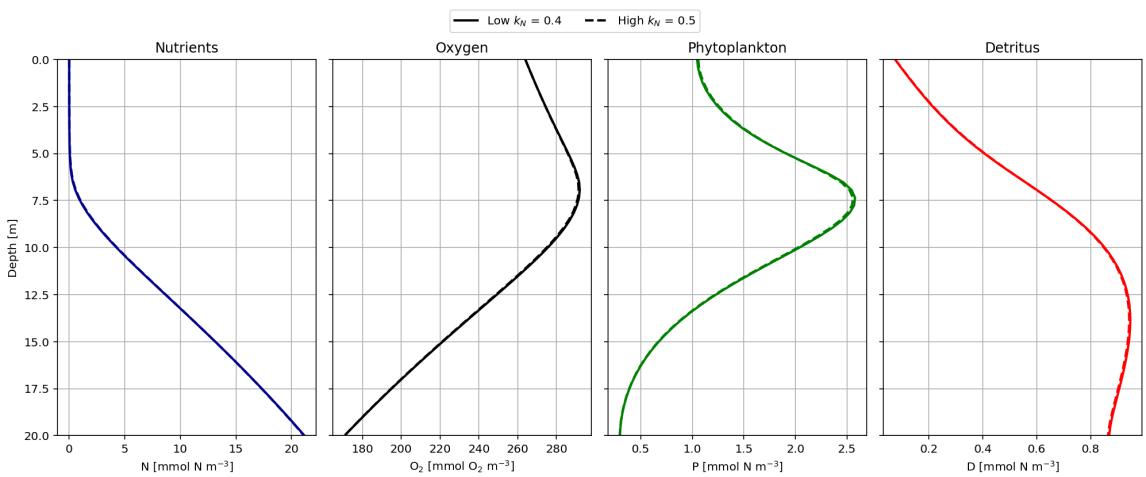

Figure 3.8: Sensitivity of the ecosystem structure to changes in the nutrient half-saturation constant.

In contrast to the mortality experiment, changes in nutrient uptake efficiency do not significantly alter the vertical position of the phytoplankton maximum. Instead, the entire biomass profile is slightly reduced while maintaining a similar depth structure.

This occurs because the location of the phytoplankton maximum is primarily determined by the balance between light and nutrient limitation, which is not strongly shifted by moderate changes in the half-saturation constant.

Overall, this relatively weak sensitivity indicates that nutrient uptake efficiency mainly affects the magnitude of phytoplankton biomass but does not fundamentally alter the vertical ecosystem structure under the simulated conditions.

3.4.3 Sensitivity to Detritus Sinking Velocity

The final experiment explores the effect of the detrital sinking velocity w_D, which controls how rapidly organic matter is transported downward through the water column. The influence of this parameter arises through the vertical transport term in the detritus equation

$$\frac{\partial D}{\partial t} = -w_D \frac{\partial D}{\partial z} + \text{sources} - \text{sinks}, \tag{3.4}$$

where w_D represents the downward sinking velocity of detrital particles. Increasing w_D strengthens the downward advective flux of detritus and therefore accelerates the export of organic material from the productive surface layer.

Mechanistically, the resulting pathway can be summarized as

$$w_D \uparrow \Rightarrow \text{ detritus export} \uparrow \Rightarrow \text{ remineralization at depth} \uparrow \Rightarrow N_{\text{deep}} \uparrow . \tag{3.5}$$

As detrital particles sink more rapidly, less detritus accumulates in the upper water column and a larger fraction of organic matter is transported to deeper layers before being remineralized. This increases the separation between surface production and deep remineralization, a process commonly associated with the biological pump.

In contrast to the mortality experiment, changes in detritus sinking velocity have only a minor effect on the vertical distribution of phytoplankton biomass. The position of the phytoplankton maximum remains approximately at the same depth because phytoplankton growth is primarily controlled by the balance between light availability and nutrient limitation rather than by detritus transport processes.

Instead, the main effect of increased sinking velocity is a redistribution of organic matter within the water column. Detritus concentrations decrease in the upper layers while remineralization becomes more pronounced at depth, leading to slightly elevated nutrient concentrations in deeper waters.

Ecologically, variations in the detritus sinking velocity can be interpreted as reflecting

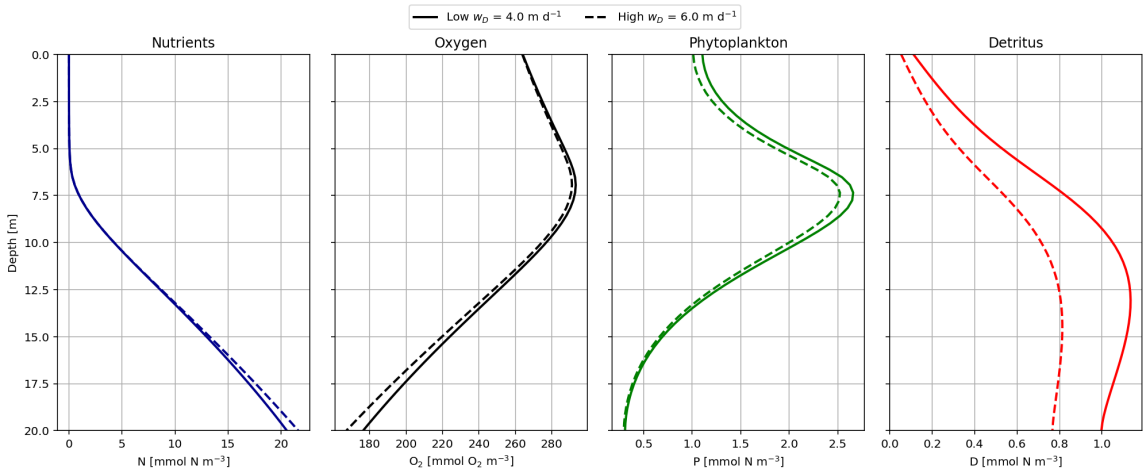

Figure 3.9: Sensitivity of the ecosystem structure to changes in detritus sinking velocity.

differences in particle size or plankton community composition. Larger phytoplankton species, such as diatoms, tend to produce heavier detritus that sinks rapidly, whereas smaller phytoplankton species produce more slowly sinking organic matter. Ecosystems dominated by large plankton therefore typically exhibit stronger vertical export of organic material and a greater separation between surface production and deep remineralization.

Conclusion

The ecosystem model developed in this study provides a simplified representation of nutrient cycling, biological production, and oxygen dynamics in a vertically resolved water column. Despite its conceptual simplicity, the model reproduces several key features of seasonal marine ecosystems. In particular, the simulations capture the typical succession of winter nutrient accumulation, a pronounced spring phytoplankton bloom, summer nutrient limitation, and oxygen depletion below the productive surface layer. The timing of the spring bloom and the general magnitude of phytoplankton biomass emerge naturally from the seasonal variations in light availability and vertical mixing, demonstrating that these physical drivers exert a strong first-order control on ecosystem dynamics.

The model further illustrates how biological production in the surface layer is linked to oxygen consumption at depth through sinking organic matter and subsequent remineralization. Sensitivity experiments highlight the importance of parameters controlling phytoplankton losses and particle export, which regulate the partitioning between living biomass and detritus and therefore influence subsurface oxygen dynamics.

At the same time, the simplified formulation imposes clear limitations. The model only considers nitrogen as the limiting nutrient and does not represent other potentially important elements such as phosphorus or sulfur, which can play a crucial role in highly productive or oxygen-depleted environments. In addition, the sediment is represented as a depth-integrated reservoir, which does not resolve the internal biogeochemical structure of benthic sediments.

Future work could therefore extend the model by introducing multiple phytoplankton functional types with different ecological traits. Distinguishing between groups such as small phytoplankton and larger, faster-sinking species would allow the model to represent differences in nutrient uptake efficiency, growth strategies, and particle export. Such an extension would improve the representation of ecosystem structure and could help explain how shifts in plankton community composition influence vertical export of organic matter and the development of oxygen depletion at depth. Nevertheless, the present model already demonstrates how a relatively simple ecosystem formulation can reproduce the main seasonal patterns of marine productivity and oxygen variability.